

Advanced (Habanero)

- Q1.** Solve for x in $x^2 + 4x + 4 = 0$ using the quadratic formula.
(A) $x = -2$
(B) $x = -4$
(C) $x = 0$
(D) $x = 2$
(E) $x = 4$
- Q2.** What is the discriminant of $x^2 - 7x + 12 = 0$?
(A) $(-7)^2 - 4(1)(12)$
(B) $(-7)^2 + 4(1)(12)$
(C) $7^2 - 4(1)(12)$
(D) $7^2 + 4(1)(12)$
(E) $(-12)^2 - 4(1)(-7)$
- Q3.** How many solutions does $x^2 + 2x + 2 = 0$ have?
(A) 0
(B) 1
(C) 2
(D) infinite
(E) cannot be determined
- Q4.** Solve for x in $x^2 - 9x + 20 = 0$ using the quadratic formula.
(A) $x = 4$ or $x = 5$
(B) $x = -4$ or $x = -5$
(C) $x = 4$ or $x = -5$
(D) $x = -4$ or $x = 5$
(E) $x = 2$ or $x = 10$
- Q5.** What is the value of the discriminant for $x^2 - 3x - 4 = 0$?
(A) $(-3)^2 - 4(1)(-4)$
(B) $(-3)^2 + 4(1)(-4)$
(C) $3^2 - 4(1)(-4)$
(D) $3^2 + 4(1)(-4)$
(E) $(-4)^2 - 4(1)(-3)$
- Q6.** How many real solutions does $x^2 + 2x + 5 = 0$ have?
(A) 0
(B) 1
(C) 2
(D) infinite
(E) cannot be determined
- Q7.** Solve for x in $x^2 + 5x + 6 = 0$ using the quadratic formula.
(A) $x = -2$ or $x = -3$
(B) $x = -2$ or $x = 3$
(C) $x = 2$ or $x = -3$
(D) $x = 2$ or $x = 3$
(E) $x = -1$ or $x = -6$
- Q8.** What is the value of the discriminant for $x^2 - 2x - 8 = 0$?
(A) $(-2)^2 - 4(1)(-8)$
(B) $(-2)^2 + 4(1)(-8)$
(C) $2^2 - 4(1)(-8)$
(D) $2^2 + 4(1)(-8)$
(E) $(-8)^2 - 4(1)(-2)$

Open Ended Questions (FRQ)

- Q9.** Solve $x^2 - 6x + 8 = 0$ using the quadratic formula and determine the number and type of solutions.
- Q10.** Given the equation $x^2 + 4x + 3 = 0$, find the value of the discriminant and interpret the results to determine the number and type of solutions.

Chef's Tips

- Ensure students show all work and explain their reasoning for free response questions.
- Remind students to check their calculations carefully.
- Allow students to use a calculator to check their work.